

The fixed-order minimum $L_5(5) = 72$

Xiaojian Zeng

PBC School of Finance, Tsinghua University, 100190

xiaojianzeng@foxmail.com

Abstract

We determine the least diameter of a five-element integer alphabet having no nonzero balanced relation of positive mass at most five. The value is $L_5(5) = 72$, attained by $(0, 1, 16, 66, 72)$. An exact certificate contains 342 572 forbidden relations on initial prefixes and covers all 971 635 alphabets of smaller diameter. We explain why the certificate checks imply complete exclusion and give a short, independent exhaustive check using only integer arithmetic. The result concerns the fixed order $q = 5$; it does not determine $L_5(q)$ for arbitrary q or classify all minimizers. Broader bibliographic priority is not asserted.

1 Statement and context

An *alphabet* is a strictly increasing tuple

$$S = (s_0, s_1, s_2, s_3, s_4), \quad 0 = s_0 < s_1 < s_2 < s_3 < s_4 = D, \quad s_i \in \mathbb{Z}.$$

For $z \in \mathbb{Z}^5$, write $m(z) = \sum_{i=0}^4 \max(z_i, 0)$. A nonzero vector z is a *balanced relation* on S if

$$\sum_{i=0}^4 z_i = 0, \quad \sum_{i=0}^4 z_i s_i = 0. \tag{1}$$

We call S admissible at order q if it has no such relation with $m(z) \leq q$, and let $L_5(q)$ be the least possible diameter of an admissible five-element alphabet. Equal sums are compared at the same length, repetitions are allowed, and reordering a multiset does not produce a different multiplicity vector.

Zhang's Question 12.2 [1] asks for $L_5(q)$ and records $L_5(2) = 11$, $L_5(3) = 23$, and $L_5(4) = 41$. The general bound there gives $L_5(5) \leq 150$; the construction displayed in Table 5 of that paper is $(0, 1, 16, 22, 111)$ at $q = 5$. The present contribution is a matching construction and exhaustive lower bound at this one order.

Theorem 1. *The exact fixed-order minimum is $L_5(5) = 72$. It is attained by*

$$A = (0, 1, 16, 66, 72).$$

Every strictly increasing five-element integer alphabet beginning at zero and ending below 72 has a nonzero balanced relation of mass at most five.

The comparison above is with the supplied primary version of [1]. Wider tables of generalized Sidon sets and related integer models have not been comprehensively checked, so no claim of publication priority is made. As adjacent computational methodology, Zeng's strict-pair product extremal problem [2] also combines an arithmetic collision-free construction with exact finite exclusion. That reference concerns products of distinct pairs and supplies no theorem about the additive minimum considered here.

2 The finite certificate proof

Lemma 2. *An alphabet S is admissible at order five if and only if the map*

$$u \mapsto \sum_{i=0}^4 u_i s_i, \quad u \in \mathbb{Z}_{\geq 0}^5, \quad \sum_{i=0}^4 u_i = 5, \quad (2)$$

is injective. A forbidden relation on an initial prefix remains forbidden on every extension of that prefix.

Proof. If distinct multiplicity vectors u, v of the same weight $h \leq 5$ give equal sums, then $z = u - v$ satisfies (1), is nonzero, and has $m(z) \leq h$. Conversely, the positive and negative parts of a nonzero balanced relation have equal weight $m(z)$ and equal sums. Adding $5 - m(z)$ zero letters to each side gives distinct vectors in the domain of (2). This also shows that checking exactly five letters checks every smaller length. A relation on a prefix extends by zero coefficients on the added coordinates, preserving balance, mass, and its vanishing sum. $\square$

2.1 The attaining alphabet

There are $\binom{9}{4} = 126$ vectors in the domain of (2). For $A = (0, 1, 16, 66, 72)$ their sums are all distinct. The supplied file `witness-72-sums.csv` lists all multiplicities and sums; checking it requires only the following domain:

$$(u_1, u_2, u_3, u_4) \in \{0, \dots, 5\}^4, \quad u_1 + u_2 + u_3 + u_4 \leq 5, \quad u_0 = 5 - u_1 - u_2 - u_3 - u_4.$$

The sum to check is $u_1 + 16u_2 + 66u_3 + 72u_4$. The checker in Appendix A verifies all 126 values without using that file. By Lemma 2, this establishes $L_5(5) \leq 72$.

As redundant checks, a separate implementation enumerated the multisets at each weight $h = 0, \dots, 5$ and found respectively 1, 5, 15, 35, 70, 126 distinct sums, exactly $\binom{h+4}{4}$ in each case. Direct enumeration also checked all 2750 nonzero balanced vectors of mass at most five, counting both signs, and found none orthogonal to A . This domain is finite because every coordinate of such a vector lies in $[-5, 5]$.

2.2 Complete exclusion below 72

Every alphabet of diameter below 72 corresponds uniquely to a four-element subset of $\{1, \dots, 71\}$. Thus the full domain is

$$\mathcal{D} = \{(0, a, b, c, d) : 1 \leq a < b < c < d \leq 71\}, \quad |\mathcal{D}| = \binom{71}{4} = 971\,635. \quad (3)$$

Diameters zero through three have no five-element alphabets. No gcd, reflection, or other symmetry restriction is imposed.

Let $P = (0, p_1, \dots, p_{k-1})$ be an initial prefix of an alphabet in $\mathcal{D}$, and let $C(P)$ denote all members of $\mathcal{D}$ that begin with P . The number of extensions is

$$|C(P)| = \binom{71 - p_{k-1}}{5 - k}, \quad (4)$$

because the remaining $5 - k$ entries are an increasing choice from the $71 - p_{k-1}$ larger integers.

Proposition 3. *Suppose a collection of distinct, extendable prefixes of $\mathcal{D}$ has the following properties: no listed prefix is a proper initial segment of another; each carries a nonzero balanced relation of mass at most five supported on that prefix; and the sum of (4) over the collection is $\binom{71}{4}$. Then every alphabet in $\mathcal{D}$ is inadmissible.*

Proof. If two prefixes have a common extension, the shorter is an initial segment of the longer. The stated condition therefore makes their extension sets pairwise disjoint. These are subsets of $\mathcal{D}$, and their cardinalities sum to $|\mathcal{D}|$, so their union is all of $\mathcal{D}$. Each member inherits a forbidden relation by Lemma 2. $\square$

The compressed certificate `exclusions-through-71.csv.gz` satisfies Proposition 3. A row records k , the prefix entries, five integer coefficients z_i , and the extension count. Unused alphabet entries are empty and unused coefficients are zero. The verifier checks strict order from zero, extendability $p_{k-1} \leq 71 - (5 - k)$, correct padding, both equalities in (1), and $1 \leq m(z) \leq 5$. It then checks distinctness and the absence of any listed proper initial segment, and sums the binomial extension counts. The resulting totals are

Prefix length	Number of certificate rows
3	166
4	25 284
5	317 122
Total	342 572

The weighted extension count is 971 635. A redundant pass enumerates every member of (3), checks that it has exactly one listed prefix, and recovers $\binom{d-1}{3}$ alphabets of each diameter $d = 4, \dots, 71$. Hence Proposition 3 gives $L_5(5) \geq 72$, completing the proof of Theorem 1.

The producer grows increasing prefixes from (0) and records a collision as soon as it occurs. A rejected prefix needs no further extension. Otherwise all permissible next entries are tried, leaving enough larger integers for a full alphabet. The proof does not depend on this search strategy: the relation identities and the complete coverage conditions are checked separately from the producer.

3 Reproduction and scope

The supplement includes the compressed certificate and the full witness table. The producer is `certify_minimum.py`; a separate checker is `verify_minimum.py`. All require only Python 3 and its standard library. From the supplement directory, run

```
python3 evidence/verify_minimum.py
python3 evidence/test_certificate_rejection.py
python3 direct_check.py
```

The first command validates every relation and all coverage and witness checks described above. The second confirms rejection of three damaged copies: a changed balanced coefficient vector, an omitted prefix, and an incorrect witness sum. These negative tests supplement the positive verification; they do not establish completeness by themselves.

The last command runs the complete direct check printed in Appendix A. It consults no certificate and uses no pruning: it enumerates each four-element subset of $\{1, \dots, 71\}$ and finds a collision among its 126 multiplicities. A separate review implementation likewise checked the entire domain without consulting the certificate, and verified its prefix coverage using contiguous lexicographic intervals. All decisive tests use exact integer arithmetic. No floating-point tolerance, probabilistic inference, solver-reported optimum, or unreported search cutoff enters the result.

The theorem resolves exactly the fixed numerical problem $L_5(5)$. The parameter-uniform question and a classification of all alphabets attaining 72 are outside this result. The bibliographic comparison is limited to the inspected source described in Section 1; broader novelty and overlap with existing tables remain unverified.

Computational and AI assistance. OpenAI Codex assisted with the search, exact-arithmetic programs, mathematical exposition, verification, and manuscript preparation. A separate AI review context implemented additional checks independently of the producer. This was a separate verification pass, not external peer review or proof-assistant formalization. The mathematical evidence is the finite computation with the stated domains and the counting argument above, rather than the identity or confidence of the assistant.

A A complete direct checker

Each tuple in U specifies the multiplicities of the four positive letters; the zero letter supplies the remaining multiplicity. Thus its domain is in bijection with (2). The function `collision` returns true exactly when that map is not injective. The outer loop is precisely (3), and the diameter counts provide an additional check. Successful execution prints the two lines shown at the end. The certificate gives a separately replayable lower bound with an explicit relation for each rejected prefix.

```
"""Complete integer check for L_5(5)=72; Python 3 standard library."""
from itertools import combinations, product
from math import comb

def require(condition, message):
    if not condition:
        raise ValueError(message)

# The missing coordinate is the multiplicity of the zero letter.
U = [u for u in product(range(6), repeat=4) if sum(u) <= 5]
require(len(U) == comb(9, 4) == 126, "multiplicity domain")

def collision(tail):
    a, b, c, d = tail
    seen = set()
    for i, j, k, l in U:
        value = i*a + j*b + k*c + l*d
        if value in seen:
            return True
        seen.add(value)
    return False

require(not collision((1, 16, 66, 72)), "attaining witness")
per_diameter = [0] * 72
for tail in combinations(range(1, 72), 4):
    require(collision(tail), ("smaller witness", tail))
    per_diameter[tail[-1]] += 1
for d in range(72):
    expected = comb(d-1, 3) if d >= 4 else 0
    require(per_diameter[d] == expected, ("diameter count", d))
require(sum(per_diameter) == comb(71, 4) == 971635,
        "complete lower-diameter domain")
print("Witness: 126 distinct sums; smaller alphabets excluded: 971635")
print("L_5(5) = 72")
```

References

- [1] Enkai Zhang, *Sharp order-preserving integer models for short additive equalities*, arXiv:2609.08915v1 [math.NT], 8 September 2026. <https://arxiv.org/abs/2609.08915>.
- [2] Zijian Zeng, *The Strict-pair Product Extremal Value at 62 is 31*, 2026. <https://doi.org/10.2139/ssrn.7380458>.